Brunda Marpadaga

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Objective

Full-time/research embedded firmware or post-silicon bring-up role starting immediately.

Education

Portland State University, Oregon **GPA: 3.85/4**

Masters in Electrical and Computer Engineering

- Coursework : Microprocessor System Design, Computer Architecture, Python Scripting, SoC Design with FPGAs, Embedded Design with FPGAs, Emulation and Functional Specification Verification, Post-Silicon Validation, Embedded Systems Programming

Work Experience

Associate Software Engineer Jul 2022 – Dec 2023

Prodapt *Hyderabad, India*

Developed full-stack web applications using Java SpringBoot and MongoDB with familiarity in Docker containers. Gained experience with version control (Git) and Agile workflows.

Failure Analysis Intern May 2019 – Dec 2019

Electronics Corporation of India Limited *Hyderabad, India*

Performed in-circuit testing, functional testing, and root-cause analysis across production stages including schematic reviews, component replacements and calibration of embedded boards for Smart Energy Meters.

Technical Skills

Languages: C, C++, SystemVerilog, Python, ARM Assembly, MATLAB
Architectures & Protocols: RISC-V, I2C, Wishbone, UART, SPI, AXI. Familiar with CAN
Operating systems: Comfortable with Windows, Linux
Hardware: Nexys A7-100t, Zybo Z7-20, Arduino, STM32
Verification & Debug: Oscilloscope, Logic Analyzer, JTAG, Multimeter, GDB, Valgrind, J-Link/Segger Ozone
EDA & Toolchains: Questasim, Vivado & Vitis, STM32CubeIDE, Cmake, Multisim, SPICE
Networking protocols: TCP/IP, MQTT
Firmware & Systems: Comfortable with reading data sheets and schematics. Knowledgeable on RTOS, inter-task communication, Multithread programming, sensor interfacing; Experienced debugging device-tree and kernel driver issues

Projects

Embedded-C FPGA based Intensity Control of an LED using TSL2561 sensor and Closed-loop control

Developed a MicroBlaze soft-core based embedded system on the Nexys-A7 using Vivado for Board Bring-up & FreeRTOS and created I2C low-level driver code for light sensor to monitor the luminescence and provided feedback through PID control loop to LED output so it could be PWM-controlled to maintain desirable lighting.

Embedded Linux Firmware bring-up: BME280 + MQTT/TLS + Ethernet on Zynq-7020 SoC

Built an embedded Linux IOT system on Zybo Z7-20 by integrating AXI GPIO & I2C peripherals in FPGA fabric; C firmware for BME280 temp/humidity sensor & OLED display. Published live sensor data as JSON to HiveMQ Cloud over MQTT via Ethernet. Cross-compiled applications & BitBake recipes using the PetaLinux Yocto build system & integrated a MATLAB live dashboard subscribing to the MQTT stream for real-time visualization.

Dual-Core AMP Benchmark: HW Accelerator vs. SW Matrix Multiply on Zynq-7020

Benchmarked a Tiled HLS matmul accelerator running on PL (Core 0, FreeRTOS) against an ARM Cortex-A9 software baseline (Core 1, bare-metal) measuring up to 4.89× speedup at N=512 matrix size.
Read per-core ARM CP15 PMU and shared PL310 L2 performance registers to characterize cache behavior under concurrent dual-core load.

ESP8266 Arduino C++ Design of Remote Monitoring System for Infants

Developed an automated IoT cradle on an ESP8266 to process telemetry from 3 sensors, using hardware interrupts for low-latency sampling. Transmitted data over Wi-Fi via HTTP to ThingSpeak for MATLAB analysis, while driving a servo-motor for physical actuation and configuring IFTTT webhooks for real-time mobile alerts.